

 Cloud Data Lake Status Sync & Ingest from AWS S3

Data Inventory Tracking

 products.csv

8 Rows

 customers.csv

10,000 Rows

 sales.csv

500,000 Rows



Retail Operations Intelligence Engine

Production-Grade Agentic Big Data Pipeline with LangGraph & MLflow



Query the Operations Analyst

 Try a sample operational query:

What was the total revenue across all transactions?

 Or enter a custom business analyst question:

What was the total revenue across all transactions?



Execute Agent Analysis



Analytical Insights

Executive Summary

Across all recorded transactions, cumulative total revenue reached 122,428,987.64 * *(approx. * \$122.43M)*.

Core Takeaways

- **Top-Line Benchmark:** Represents the aggregate gross transactional value across all sales channels, providing a critical baseline for top-line performance.
- **Scale of Operations:** Reflects strong overall transaction volume and commercial activity across the business portfolio.

Strategic Recommendation

To maximize the strategic value of this metric, recommend conducting a secondary analysis to evaluate:

1. **Revenue Drivers:** Breakdown by product line, region, and customer segment.
2. **Unit Economics:** Average Order Value (AOV) and customer lifetime value (LTV).

3. Period-over-Period Growth: Growth trends compared to previous operating cycles.

 Self-Correction Retries Required

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 Execution Code Trace

Executed Pandas Data Manipulation Pipeline Code:

```
import pandas as pd

try:
    sales_df = sales
except NameError:
    sales_df = pd.read_csv("sales.csv")

result = sales_df["Total_Revenue"].sum()
```

Calculated Raw Context Outputs:

122428987.64

 View Live System Graph Architecture

